

KLOTEN, Sweden

WMO: 024310

Lat: 59.8703N

Lon: 15.2522E

Elev: 948

StdP: 14.20

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.2	5.1	-6.1	4.1	1.3	0.7	5.9	6.1	16.2	32.2	14.4	31.8	1.8	30	0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	77.8	62.4	74.5	61.0	71.2	59.5	65.3	73.0	63.4	70.5	61.7	68.2	5.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.8	88.5	66.5	61.0	82.9	65.1	59.0	77.2	64.3	30.6	73.5	29.3	70.7	28.0	67.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.9	12.2	11.0	-8.8	81.6	8.1	3.1	-14.6	83.9	-19.4	85.7	-23.9	87.5	-29.8	89.8		
			-8.4	67.3	8.0	2.8	-14.2	69.3	-18.9	71.0	-23.4	72.6	-29.2	74.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	22.5	24.5	30.5	39.0	49.5	56.0	61.1	57.9	51.1	40.5	33.1	26.9
	DBStd	14.90	8.73	8.97	8.03	6.59	7.73	5.30	5.42	4.50	4.85	5.80	6.72	9.18
	HDD50	4188	852	715	605	334	106	9	0	1	44	299	506	717
	HDD65	8750	1317	1135	1070	780	482	275	149	227	418	759	956	1182
	CDD50	953	0	0	0	4	90	189	344	245	76	5	0	0
	CDD65	39	0	0	0	0	0	5	28	6	0	0	0	0
	CDH74	328	0	0	0	0	18	48	225	37	0	0	0	0
	CDH80	35	0	0	0	0	0	1	29	5	0	0	0	0
	WSAvg	5.2	5.1	5.3	5.6	5.3	5.5	5.3	4.9	4.6	5.2	5.4	5.5	5.3
	PrecAvg	28.7	2.2	1.6	1.3	1.7	2.3	2.8	3.3	3.3	2.4	2.8	2.6	2.4
Precipitation	PrecMax	40.5	4.2	3.6	3.0	5.0	5.4	5.8	7.5	7.4	4.9	7.2	6.7	4.3
	PrecMin	20.9	0.2	0.4	0.0	0.0	0.7	0.1	0.5	0.9	0.4	0.0	0.8	0.6
	PrecStd	3.8	1.0	0.8	0.8	1.1	1.1	1.4	1.3	1.4	1.5	1.7	1.3	1.2
	PrecStd	3.8	1.0	0.8	0.8	1.1	1.1	1.4	1.3	1.4	1.5	1.7	1.3	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.9	44.4	56.1	65.8	76.6	79.3	83.3	78.6	69.1	59.0	49.1	46.7
		MCWB	40.3	37.7	46.2	49.8	57.7	61.1	64.2	64.4	58.9	53.2	48.0	45.8
	2%	DB	38.3	39.7	48.6	61.4	72.5	75.0	79.6	73.2	64.7	54.7	46.1	44.1
		MCWB	37.7	37.0	41.3	47.9	56.3	60.5	64.2	62.4	57.5	51.9	45.4	43.2
	5%	DB	36.0	37.3	45.0	55.8	68.8	71.1	76.2	69.7	61.6	51.9	44.0	40.7
		MCWB	35.3	35.6	38.3	45.8	55.2	58.7	62.7	60.0	56.1	50.1	43.4	39.7
	10%	DB	33.7	35.5	41.4	51.0	64.1	67.0	72.7	66.8	59.1	49.6	42.4	37.9
		MCWB	33.0	34.1	36.4	42.6	53.7	56.1	61.2	58.8	54.3	47.4	42.0	36.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.6	39.8	46.0	51.3	60.8	64.3	68.3	66.5	61.6	55.0	48.0	46.3
		MCDB	40.6	41.0	55.5	63.7	70.3	72.4	75.8	73.3	65.4	57.0	48.1	46.5
	2%	WB	37.8	37.6	42.1	48.8	58.8	62.2	66.3	64.1	59.1	53.0	45.5	43.3
		MCDB	38.1	38.9	47.7	59.0	68.0	70.3	74.6	69.9	63.4	54.2	46.0	44.0
	5%	WB	35.6	36.0	39.6	46.5	56.8	60.2	64.4	62.2	57.2	50.6	43.7	39.8
		MCDB	36.2	37.2	43.3	54.1	65.4	68.5	72.6	66.9	59.9	51.7	44.0	40.4
	10%	WB	33.4	34.1	37.2	44.0	54.5	58.0	62.8	60.6	55.5	47.8	42.0	37.0
		MCDB	33.8	35.2	40.2	49.4	62.5	65.4	70.5	64.7	58.0	48.8	42.4	37.6
Mean Daily Temperature Range	5% DB	MDBR	8.7	10.0	14.2	16.9	18.0	16.5	16.8	15.1	13.0	10.0	6.8	8.2
		MCDBR	7.9	11.8	17.7	25.0	24.7	23.1	22.3	20.0	16.6	12.1	8.7	9.2
	5% WB	MCWBR	7.8	8.9	11.2	13.5	12.2	11.1	9.1	10.1	9.7	9.2	8.1	9.8
		MCDWR	8.1	9.3	16.3	22.0	21.3	19.6	19.4	16.6	14.5	10.9	8.6	9.5
	5% WB	MCWBR	8.2	7.2	11.2	12.7	11.5	10.4	8.8	9.7	9.8	9.2	8.2	10.1
		MCWBR	8.2	7.2	11.2	12.7	11.5	10.4	8.8	9.7	9.8	9.2	8.2	10.1
Clear-Sky Solar Irradiance	taub		0.255	0.298	0.321	0.357	0.355	0.347	0.368	0.361	0.329	0.317	0.279	0.234
	taud		2.216	2.313	2.381	2.367	2.431	2.473	2.429	2.444	2.525	2.468	2.283	2.120
	Ebn at Noon		172	226	259	268	277	281	271	264	256	220	160	129
	Edh at Noon		13	21	26	32	32	31	32	30	23	18	12	9
All-Sky Solar Radiation	RadAvg		93	290	777	1228	1542	1754	1594	1248	827	371	118	61
	RadStd		16	54	92	143	151	117	188	137	109	67	30	6

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-214	N/A	N/A	N/A

Nomenclature: See separate page